ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Rockwell/Allen-Bradley Stratix 2000 Unmanaged Industrial Ethernet Switch (1783-US series) · Component type: **Network Device (NDR) - UNMANAGED Layer 2 industrial Ethernet switch** · Target SL = SL 3

Where to work	There is **NO management interface** - no CLI, no web, no SSH, no SNMP, no IP address, no configuration. The only on-device items are the **front-panel status LEDs** (diagnostics) and a **top-panel DIP switch** for broadcast-storm protection (enable it). Assess the device by **physical inspection** plus the **UPSTREAM managed switch** that connects it - that switch must enforce segmentation, port-security/802.1X, and monitoring for the Stratix 2000's segment. Grounded in Rockwell **1783-UM011** (Stratix 2000 Unmanaged Switches).
Ports / paths	**Ethernet only (Layer 2)**: 10/100 (or Gigabit) RJ45 copper + optional fiber SFP; DIN-rail; power **12-48 VDC / 18-30 VAC**. **NO management ports** (no console / USB / SD), **NO IP**, no PTP. Top-panel **DIP switch = broadcast-storm protection**; front-panel **LEDs = diagnostics**. Catalogs e.g. **1783-US5T / US8T / US7T1F / US4T1F / US8TG2GX**.
Notes / lifecycle	UNMANAGED switch - it provides **NO NDR security controls** (no I&A, no VLAN/port-security, no logging, no crypto). It **cannot meet SL-3 natively**: use it only within a **single small trusted microsegment** behind a managed switch, with physical control + upstream enforcement + off-device monitoring, and **plan replacement with a managed Stratix (5700/5800)** for SL-3 zones. Track Rockwell PSA / CISA advisories. Nearly every row below is N-A-at-device + Compensating.
CLI syntax note	The Stratix 2000 has **no CLI** — all command-line verification happens on the **upstream managed switch**, so use that switch's syntax: a **Stratix 5700 / 5400 is classic IOS** (`show authentication sessions`, `show macsec interface <if>`) while a **Stratix 5800 / 5200 is IOS-XE** (`show access-session`, `show macsec summary`) — confirm with `show version` before running the upstream checks (see those sheets).
How to read	Default = physical inspection + review of the UPSTREAM managed switch (the Stratix 2000 cannot be logged into or scanned for management). Each row states where the control actually lives and how to verify the compensating control. Test/exploit tools target the segment/upstream device, not the unmanaged switch. Most rows are honestly N-A-at-device / Compensating (upstream + physical).

Session setup & evidence handling

Do this BEFORE running any command so the evidence is complete, attributable, and protected:
1. Start a full session log first. PuTTY: Session › Logging → select **All session output** → set **Log file name** (browse to the encrypted evidence folder) → under "if the log file already exists" pick **Always append** → tick **Include header** and (Logging › Options) **Flush log frequently**; put `&Y&M&D_&T` in the name for an auto timestamp. (SecureCRT: File › Log Session; Tera Term: File › Log; macOS/Linux: `script -a <file>.log` or `ssh user@host | tee <file>.log`.)
2. Name the file to the evidence convention (one file per device): `<site>_<zone>_<assetTag>_<vendor>_<model>_<mgmtIP>_<YYYYMMDD-HHMM>_<testerInitials>.log`.
3. Disable paging so full output is captured (it resets each login): this device has **no management CLI** - do not try to log into it; capture is physical inspection plus the **UPSTREAM managed switch** session (log that per its own sheet).
4. Self-document the log - record the device identity from the label/asset tag and photograph the install, so the file proves what/when/who.
5. Close out: restore paging, stop the log (PuTTY closes it when the session ends), and verify the file opened/saved.
6. Protect the evidence file. It contains running-config and may hold hashes, SNMP community strings, or keys: store **only** on the encrypted assessment share, follow the project data-classification/retention policy, redact secrets before sharing, and **never** email it in cleartext. **Do not save the device password in the PuTTY/SecureCRT saved session** - authenticate interactively or via the approved password manager.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 1.1	Human user I&A	**N/A at device** - the Stratix 2000 has no management interface or accounts to authenticate (nothing to log into). This removes a remote-mgmt attack surface but also means **no device-level I&A**. Confirm the switch truly exposes no management, and that admin access to the **upstream managed switch** (which governs this segment) is authenticated (AAA). **Test:** scan the Stratix 2000 - it has no IP / no mgmt ports answer (nothing to authenticate); confirm the upstream switch requires login. **Ext. tools:** `nmap -Pn -p 22,23,80,443,161 <sw-segment>` (no mgmt on the unmanaged switch); physical confirm no console/mgmt port; review upstream AAA.	No device mgmt to authenticate (N/A); upstream managed switch authenticates admin access.	N/A at device - Compensating.
SR/CR 1.2	Device & software-process I&A	**N/A at device** - the Stratix 2000 has **no 802.1X / MAB / MAC filtering**; any device plugged into any port gets Layer-2 connectivity. The control must be UPSTREAM: the managed-switch port that uplinks the Stratix 2000 enforces 802.1X/MAB or a MAC limit, and only a small set of known, physically-controlled devices connect to the unmanaged switch. **Test:** plug a rogue device into a Stratix 2000 port -> it gets L2 connectivity (the switch cannot stop it); confirm the upstream port-security/802.1X + segmentation contain the blast radius. **Ext. tools:** Rogue-device test on a port; upstream `show authentication sessions` / port-security; Wireshark on the uplink.	No device port-auth (N/A); upstream 802.1X/port-security + limited trusted endpoints compensate.	N/A at device - Compensating.
SR/CR 1.3	Account management	**N/A.** No accounts on the device. Manage admin identities on the upstream managed switch (named, least-privilege, AAA) and keep a record of who may physically access the cabinet. **Ext. tools:** Upstream AAA account review; physical-access record.	No device accounts (N/A); upstream identities managed.	N/A.
SR/CR 1.4	Identifier management	The device has a fixed unique **MAC / serial** (nameplate) but no configurable identity. Inventory each unit (catalog, serial, MAC, location, transformer/cell served) so it is uniquely tracked. **Ext. tools:** Nameplate (catalog/serial/MAC); asset register.	Unique hardware identity inventoried; no configurable identity to manage.	Inventory only.
SR/CR 1.5	Authenticator management	**N/A.** The device stores no credentials/keys. Authenticator management applies only to the upstream managed switch. **Ext. tools:** Upstream credential/key review.	No device credentials (N/A); upstream authenticators managed.	N/A.
SR/NDR 1.6	Wireless access management	**N/A.** Wired switch with no wireless interface. **Ext. tools:** Confirm no WLAN on the platform.	N/A - justification recorded (wired device).	N/A
SR/CR 1.7	Strength of password auth	**N/A.** No passwords on the device. Password strength is enforced at the upstream managed switch / AAA. **Ext. tools:** Upstream password policy review.	N/A at device; upstream policy enforced.	N/A.
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A at device.** No management, no crypto, no certificates on an unmanaged switch. Device-identity/PKI controls live on the upstream managed switch (SUDI/certs) and endpoints. **Ext. tools:** Confirm no device certificate/crypto; review upstream PKI.	N/A device PKI; upstream/endpoint PKI applies.	N/A.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	**N/A.** No authenticator entry on the device. **Ext. tools:** N/A - confirm no login interface.	N/A - no device login.	N/A.
SR/CR 1.11	Unsuccessful login attempts	**N/A.** No login exists on the device. Lockout applies at the upstream managed switch. **Ext. tools:** Upstream login-lockout review.	N/A at device; upstream lockout enforced.	N/A.
SR/CR 1.12	System use notification	**N/A at device** (no login/display banner). Apply an approved-use placard on the cabinet and enforce the banner on the upstream managed switch (compensating). **Ext. tools:** Physical placard review; upstream banner check.	Use notice via cabinet placard + upstream banner (compensating).	Compensating.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/NDR 1.13	Access via untrusted networks	The Stratix 2000 has no management to reach, but it also **cannot restrict data flow** - it forwards all Layer-2 frames among its ports. Ensure **all of its ports sit within one trusted microsegment** and it does **not** bridge a trusted and an untrusted network; the upstream managed switch/firewall enforces the zone boundary. **Test:** confirm no untrusted device shares the unmanaged switch; from an untrusted VLAN there is no path through it into the trusted segment. **Ext. tools:** Topology/patch review (all ports same zone); `nmap -sT -p- <trusted-host>` from the untrusted side (no path via the unmanaged switch).	Unmanaged switch confined to one trusted microsegment; boundary enforced upstream.	Compensating.

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 2.1	Authorization enforcement	**N/A at device** - no roles/authorization; the switch forwards all frames among its ports without policy. Authorization (which flows are allowed) is enforced by the **upstream managed switch** (VLAN/ACL) and by physically controlling what connects. **Ext. tools:** Upstream VLAN/ACL review; physical port-usage review.	No device authorization (N/A); enforced upstream + physically.	N/A at device - Compensating.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR 2.3	Use Control for Portable and Mobile Devices	**N/A.** Device does not host portable/mobile-media use; USB/portable-device use is controlled at the engineering host (EWS).	N/A - justification recorded (not assessed as a control).	N/A
SR/NDR 2.4	Mobile code	**N/A.** Fixed-function switching ASIC - no CPU/OS running apps, no code-load path; nothing to execute mobile code. **Ext. tools:** Confirm fixed-function device (no app-hosting/code path).	N/A - no mobile-code capability (fixed-function ASIC).	N/A.
SR/CR 2.5	Session lock / timeout	**N/A.** No management session exists to lock. Applies to the upstream managed switch. **Ext. tools:** Upstream idle-timeout review.	N/A at device; upstream session lock applies.	N/A.
SR/CR 2.6	Remote session termination	**N/A.** No remote sessions to the device. Applies upstream. **Ext. tools:** Upstream session review.	N/A at device.	N/A.
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	**N/A at device** - no logging. Capture link up/down and traffic for this segment at the **upstream managed switch** (its uplink-port logs) and a passive OT IDS; there is no on-device audit trail. **Ext. tools:** Upstream `show logging` (uplink port); passive OT IDS on the segment; SIEM.	No device audit (N/A); events captured upstream/off-device.	N/A at device - Compensating.
SR/CR 2.9	Audit storage capacity	**N/A at device** - logging is entirely off-device. Confirm the upstream switch / SIEM are sized/retained and alert on collection failure so segment events are not silently lost. **Ext. tools:** Upstream/SIEM retention + source-silent alert.	Off-device logs sized/retained; audit-failure alerting defined.	Compensating.
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	**N/A.** No clock or logs on the device. Time/correlation is provided by the upstream switch / SIEM. **Ext. tools:** Upstream/SIEM time-sync review.	N/A at device; time correlation upstream.	N/A.
SR/CR 2.12	Non-repudiation	**N/A at device** - no logging/attribution. Attribution for changes to the segment is at the upstream managed switch + physical-access records. **Ext. tools:** Upstream command accounting; cabinet-access log.	N/A at device; attribution upstream/physical.	N/A at device - Compensating.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
NDR 2.13	Physical & diagnostic interfaces	**PRIMARY control.** The Ethernet ports, the top-panel **DIP switch**, and the power terminals are the only interfaces. Keep the device in a **locked cabinet**; physically block/cover **unused ports** (port blockers); protect and record the DIP-switch (broadcast-storm) setting; front-panel LEDs are the only diagnostics. **Ext. tools:** Physical inspection (cabinet lock/seal, port blockers on unused ports, DIP-switch position recorded).	Ports + DIP switch physically controlled; unused ports blocked; cabinet locked.	NDR-specific - primary.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 3.1	Communication integrity	**N/A at device** - the switch forwards Layer-2 frames in cleartext with **no integrity protection** (no MACsec). Compensating: apply **MACsec / encryption + integrity at the upstream managed switch** and endpoints, keep the unmanaged switch's segment small and physically controlled, and contain any cleartext protocols in-zone. **Ext. tools:** Wireshark on the segment (cleartext L2, contained in-zone); confirm upstream MACsec/segmentation on the uplink.	No device integrity (N/A); MACsec/segmentation upstream + physical containment.	N/A at device - Compensating.
SR/NDR 3.2	Protection from malicious code	**N/A.** Fixed-function switching ASIC with no firmware-update path and no code-execution surface - there is no software to infect and no field firmware to sign. **Ext. tools:** Confirm no firmware-update/code path (unmanaged, fixed-function).	N/A - no software attack surface (fixed-function).	N/A.
SR/CR 3.3	Security functionality verification	The only "configuration" is the **DIP-switch position** and **which ports are patched** - there is no software config to drift. Baseline it physically (photograph the DIP switch + record the port/patch map) and re-verify on inspection. **Ext. tools:** Photo/record of DIP-switch + port map as the baseline; periodic physical re-verify.	Physical config (DIP switch + port map) baselined and re-verified; no software drift.	Physical baseline.
SR/CR 3.4	Software & information integrity	Confirm software & information (config/program) integrity; unauthorized change prevented/detected. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Software & information integrity verified individually.	
SR/CR 3.5	Input validation	The ASIC forwards/floods frames and has **no management control-plane to crash** via malformed mgmt frames. The relevant integrity control is **broadcast-storm protection** (top-panel DIP switch) - enable it so a broadcast storm is limited. **ISOLATED:** generate a broadcast storm and confirm it is mitigated. **Ext. tools:** ISOLATED broadcast-storm generator; confirm the DIP-switch storm protection limits it; observe LEDs/throughput.	Broadcast-storm protection enabled and effective; no mgmt control-plane to fault.	DIP-switch control.
SR/CR 3.6	Deterministic output	**N/A.** A switch does not drive a physical process output. **Ext. tools:** N/A - network device.	N/A - justification recorded (network device).	N/A
SR/CR 3.7	Error handling	Confirm graceful link/port error behavior via the **front-panel LEDs** (link/activity/fault); there is no CPU/OS to crash or leak information. **Ext. tools:** Physical LED review; confirm defined link-fault indication.	Graceful port/link error indication; no info leak.	
SR/CR 3.8	Session integrity	**N/A.** No management sessions on the device. **Ext. tools:** N/A - no sessions.	N/A - no device sessions.	N/A.
SR/CR 3.9	Protection of audit information	**N/A at device** (no logs). Protect the upstream/SIEM store and restrict who can clear it. **Ext. tools:** Upstream/SIEM access review.	N/A at device; off-device audit protected.	N/A at device - Compensating.
NDR 3.10	Support for updates	**N/A.** Unmanaged fixed-function switch - **no field firmware update**. Record N/A; manage risk via physical control, upstream enforcement, monitoring, and a replacement plan. **Ext. tools:** Confirm no field-update path; record replacement plan.	No field update (N/A); compensating controls + replacement plan recorded.	NDR-specific / N/A.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
NDR 3.11	Physical tamper resistance/detection	**Primary control.** Keep the device in a **locked/sealed cabinet**; use port blockers on unused ports; seal where required; tamper (added/moved cables, DIP-switch change) is detected by physical inspection since there is no log. **Ext. tools:** Physical inspection (locks/seals, port blockers, DIP-switch + patch map vs baseline).	Physical access controlled; tamper detected by inspection.	NDR-specific - primary.
NDR 3.12	Provisioning product-supplier roots of trust	**N/A.** No signed firmware / supplier root of trust on this device. Record N/A; rely on hardware provenance (buy from authorized channel) + physical control. **Ext. tools:** Confirm no RoT; unit provenance from authorized channel.	No supplier RoT (N/A); provenance + physical compensate.	NDR-specific / N/A.
NDR 3.13	Provisioning asset-owner roots of trust	**N/A.** No certificate/trust provisioning (no management). Asset-owner trust is enforced at the upstream managed switch and endpoints. **Ext. tools:** Confirm no cert provisioning; review upstream trust.	No asset-owner RoT (N/A); enforced upstream.	NDR-specific / N/A.
NDR 3.14	Integrity of the boot process	**N/A.** No secure/verified boot on a fixed-function unmanaged switch. Record N/A; compensate physically + upstream + replacement plan. **Ext. tools:** Confirm no boot-integrity capability.	No verified boot (N/A); compensating controls recorded.	NDR-specific / N/A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 4.1	Information confidentiality	**N/A at device** - no cryptography; cleartext Layer-2 forwarding. Confidentiality is provided by **MACsec/segmentation at the upstream managed switch** and by keeping the segment small and physically controlled. **Ext. tools:** Confirm no device crypto; verify upstream MACsec/segmentation.	No device crypto (N/A); confidentiality upstream + segmentation.	N/A at device - Compensating.
SR/CR 4.2	Information persistence	The device stores **no configuration, credentials, or data** (only the hardware DIP-switch position). On decommission there is nothing to wipe; reset the DIP switch if desired. No sensitive data persists. **Ext. tools:** Confirm no stored config/credentials; reset DIP switch on disposal.	No sensitive data to persist; nothing recoverable on disposal.	
SR/CR 4.3	Use of cryptography	Confirm use of modern cryptography (strong suites, managed keys; weak/legacy disabled). **Steps:** for this product this is assessed with the same evidence as **SR/CR 4.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Use of cryptography verified individually.	

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 5.1	Network segmentation	**Key limitation.** An unmanaged switch **cannot segment** - no VLANs; it floods within one broadcast domain, so all its ports are a single segment. It must therefore sit **entirely within one zone/microsegment**; the **upstream managed switch** provides the VLAN/segmentation boundary. Do not use the Stratix 2000 to bridge zones. **Test:** confirm every Stratix 2000 port is in the same zone; there is no cross-zone path through it. **Ext. tools:** Patch/topology review; `nmap -sT -p- <host-in-other-zone>` from a port (no cross-zone reach via the unmanaged switch); upstream VLAN review.	Unmanaged switch wholly within one zone; segmentation boundary enforced upstream.	Compensating.
SR/NDR 5.2	Zone boundary protection	**N/A at device** - it provides **no boundary enforcement** (no ACL/firewall/CoPP). Confirm the Stratix 2000 is **not** positioned as a zone boundary; the boundary is the upstream managed switch/firewall. **Ext. tools:** Topology review (device not a boundary); upstream ACL/firewall review.	Device is not a zone boundary (N/A); boundary enforced upstream.	N/A at device - Compensating.
SR/NDR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Network switch with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services on the device.	N/A - justification recorded.	N/A

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR 5.4	Application Partitioning	**N/A.** Single-purpose device; no separable applications to partition (handled at the host/SCADA layer).	N/A - justification recorded (not assessed as a control).	N/A

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 6.1	Audit log accessibility	**N/A at device** - no logs. The segment's events are visible at the **upstream managed switch** (uplink-port logs) - confirm those are accessible/exportable to authorized staff. **Ext. tools:** Upstream `show logging` / log export.	No device logs (N/A); segment events accessible upstream.	N/A at device - Compensating.
SR/CR 6.2	Continuous monitoring	Monitor off-device: a passive **OT IDS** on a SPAN of the upstream managed-switch port (or a tap) gives visibility into traffic to/from the Stratix 2000's segment; the front-panel **LEDs** are local diagnostics only. **Ext. tools:** Passive OT IDS on the upstream SPAN/tap; SIEM; LED check for local link/fault.	Segment monitored off-device via the upstream switch; LEDs for local diagnostics.	Compensating.

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 7.1	Denial-of-service protection	Enable the on-device **broadcast-storm protection** (top-panel DIP switch) - the one availability control the device offers. Provide storm-control / DHCP-snooping / DAI / CoPP at the **upstream managed switch** for the segment. **Avoid loops**: an unmanaged switch cannot be configured for STP behaviour, so a cabling loop can melt the segment - verify no loops. **Test:** ISOLATED broadcast storm -> DIP-switch protection limits it; confirm no cabling loop exists. **Ext. tools:** ISOLATED storm generator; DIP-switch verify; loop/topology inspection; upstream storm-control review.	Broadcast-storm DIP protection on; upstream storm-control active; no loops.	DIP + upstream.
SR/CR 7.2	Resource management	The switching ASIC forwards at line rate with no configurable resources; confirm the segment is not oversubscribed and there is no CPU/table to exhaust (no management plane). **Ext. tools:** Segment throughput/oversubscription review; upstream port counters.	Line-rate forwarding; no exhaustible device resources; segment not oversubscribed.	
SR/CR 7.3	Control system backup	There is **no software config to back up** - record the model, DIP-switch setting, and port/patch map so a replacement can be swapped in immediately; hold a **spare** (swap is fast since there is no config to restore). **Ext. tools:** Config sheet (model/DIP/port map); spare unit on hand.	Physical config recorded; spare enables fast swap (no config to restore).	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security config settings	Document the **DIP-switch setting**, the port/patch usage, and the fact that the device is **unmanaged** with no security features; baseline this physically and change-control it. **Ext. tools:** Physical baseline (DIP/ports); change-control record.	Physical settings documented, baselined, change-controlled.	
SR/CR 7.7	Least functionality	The device has no services to disable (that is its nature). Least functionality here = **block/cover unused ports**, enable broadcast-storm protection, and limit connected devices to the required trusted set. **Ext. tools:** Physical inspection (unused ports blocked; DIP storm-protection on; connected-device list).	Unused ports blocked; storm protection on; only required devices connected.	
SR/CR 7.8	Component inventory	Inventory the model (1783-US catalog), serial, MAC, location/cell, and flag it as **UNMANAGED with no security features** so it feeds the SL-3 gap analysis and the managed-switch replacement decision. **Ext. tools:** Nameplate (catalog/serial/MAC); asset register (mark unmanaged / SL-3 gap); Rockwell PSA / CISA check.	Accurate inventory incl. unmanaged/SL-3-gap flag; replacement planned; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
Physical inspection / 1783-UM011	Native review - the only on-device controls (2.13, 3.11, 7.x)	Verify cabinet lock/seal, port blockers on unused ports, the top-panel DIP-switch (broadcast-storm) setting, patch/port map, and front-panel LED status against the Stratix 2000 user manual.	Read-only; record baseline.
Upstream managed switch (CLI/WebUI)	Where the real NDR controls live (most FRs)	On the managed switch feeding the Stratix 2000, verify segmentation/VLAN, port-security/802.1X on the uplink, logging, and monitoring for this segment.	Read-only; authorized.
nmap	Confirm no device mgmt + segmentation (1.1, 1.13, 5.x)	`nmap -Pn -p 22,23,80,443,161 <segment>` (the unmanaged switch has no IP/mgmt); `nmap -sT -p- <other-zone-host>` from a port to prove no cross-zone path via it.	Read-only; authorized.
Rogue-device / port test	Port-authentication compensating control (1.2)	Plug an unauthorized device into a Stratix 2000 port; confirm the UPSTREAM 802.1X/port-security + segmentation contain it (the unmanaged switch cannot).	Coordinate with NAC team.
Wireshark / tcpdump	Comms visibility & containment (3.1, 5.1, 6.2)	SPAN/tap on the upstream uplink; confirm cleartext L2 stays in-zone; use for segment monitoring (the device has no logs).	Passive; SPAN/tap only.
Broadcast-storm generator (isolated)	Storm-protection & DoS resilience (3.5, 7.1)	ISOLATED: generate a broadcast storm; confirm the DIP-switch broadcast-storm protection limits it and the segment stays available.	ISOLATED; authorization.
Passive OT IDS + SIEM	Off-device monitoring (2.8, 6.x)	On the upstream SPAN, capture segment traffic and anomalies (the Stratix 2000 has no native logging); forward to the SIEM.	Passive; coordinate with SOC.
Rockwell PSA / CISA advisories + asset register	Lifecycle & SL-3 gap (7.8)	Record the device as unmanaged / SL-3 gap; drive the managed-switch (Stratix 5700/5800) replacement decision; track advisories.	Reference.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.